

Data-prep-for-TS-worksheet

Using Claude to help find some AMT profiles for TS worksheet. We'll identify appropriate CTD casts, and make a csv file containing our selected profiles.

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.2      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
# Point this at wherever you unzipped the NCEI download
parent_dir <- "C:/Users/cmitchell/Documents/SeaChangeSemester/2026/RLabs/data/"
data_dir <- paste0(parent_dir, "0044519")
```

What's in the CSV? We know it's come from NCEI and the format these files generally are:

```
# List everything so you can see how it's organized (per-cast files vs one big file)
list.files(data_dir, recursive = TRUE) %>% head(20)
```

```
## [1] "0044519.1.1.xml"          "1.1/about/AMT data (fwd).txt"
## [3] "1.1/about/journal.txt"    "1.1/data/0-data/amt_bodc/502001.csv"
## [5] "1.1/data/0-data/amt_bodc/502001.txt" "1.1/data/0-data/amt_bodc/502002.csv"
## [7] "1.1/data/0-data/amt_bodc/502002.txt" "1.1/data/0-data/amt_bodc/502003.csv"
## [9] "1.1/data/0-data/amt_bodc/502003.txt" "1.1/data/0-data/amt_bodc/502004.csv"
## [11] "1.1/data/0-data/amt_bodc/502004.txt" "1.1/data/0-data/amt_bodc/502005.csv"
## [13] "1.1/data/0-data/amt_bodc/502005.txt" "1.1/data/0-data/amt_bodc/502006.csv"
## [15] "1.1/data/0-data/amt_bodc/502006.txt" "1.1/data/0-data/amt_bodc/502007.csv"
## [17] "1.1/data/0-data/amt_bodc/502007.txt" "1.1/data/0-data/amt_bodc/502008.csv"
## [19] "1.1/data/0-data/amt_bodc/502008.txt" "1.1/data/0-data/amt_bodc/502009.csv"
```

```
# Peek at one file's raw structure before trying to parse it
read_lines(list.files(data_dir, pattern = "\\*.csv$", full.names = TRUE, recursive = TRUE)[1],
           n_max = 15)
```

```
## [1] "Cruise, Site, Orig_Id, BODC_Id, Gear, Latitude, Longitude, Depth, Start_Date      "
## [2] "AMT6,,CTD48,502001,CTD,36 36 50N,017 30 18W,4997,10/06/1998 09:48:38              "
## [3] "PRESRP01,,ATTNMS01,,CPHLPR01,,FVLTS01,,IRRDPP01,,LVLTB01,,POTMCV01,,PSALST01,,SIGTPR01,,TEMP"
```

```
## [4] "decibars,F,per metre,F,mg/m**3,F,V,F,uE/m2/s,F,V,F,deg C,F,PSU,F,Kg/m**3,F,deg C,F"
## [5] " 2.50, , 0.164, , 0.06, , 0.0945, , 1833.52, , 2.8600, , 18.881, , 36.378, , 26.1"
## [6] " 3.50, , 0.169, , 0.06, , 0.0940, , 1533.04, , 2.7790, , 18.881, , 36.378, , 26.1"
## [7] " 4.50, , 0.165, , 0.06, , 0.0940, , 1351.23, , 2.7255, , 18.880, , 36.378, , 26.1"
## [8] " 5.50, , 0.165, , 0.06, , 0.0938, , 1266.88, , 2.6988, , 18.874, , 36.378, , 26.1"
## [9] " 6.50, , 0.165, , 0.06, , 0.0940, , 1216.18, , 2.6795, , 18.862, , 36.378, , 26.1"
## [10] " 7.50, , 0.164, , 0.06, , 0.0945, , 1108.29, , 2.6403, , 18.866, , 36.378, , 26.1"
## [11] " 8.50, , 0.165, , 0.06, , 0.0940, , 998.77, , 2.5963, , 18.858, , 36.378, , 26.1"
## [12] " 9.50, , 0.165, , 0.06, , 0.0947, , 875.41, , 2.5390, , 18.857, , 36.379, , 26.1"
## [13] " 10.50, , 0.165, , 0.06, , 0.0950, , 881.53, , 2.5420, , 18.856, , 36.379, , 26.1"
## [14] " 11.50, , 0.165, , 0.07, , 0.0957, , 720.84, , 2.4543, , 18.854, , 36.379, , 26.1"
## [15] " 12.50, , 0.164, , 0.07, , 0.0967, , 678.62, , 2.4285, , 18.848, , 36.380, , 26.1"
```

OK - from this output, we can write functions to read in the data:

```
parse_dms <- function(x) {
  # e.g. "36 36 50N" or "017 30 18W" -> decimal degrees
  hemi <- str_sub(x, -1)
  nums <- as.numeric(str_extract_all(x, "\\d+\\.?\\d*")[[1]])
  dd <- nums[1] + nums[2] / 60 + ifelse(length(nums) >= 3, nums[3], 0) / 3600
  if (hemi %in% c("S", "W")) dd <- -dd
  dd
}

read_amt_cast <- function(path) {
  raw <- read_lines(path)

  meta_names <- str_split(raw[1], ",")[[1]] %>% str_trim()
  meta_vals <- str_split(raw[2], ",")[[1]] %>% str_trim()
  meta <- set_names(as.list(meta_vals), meta_names)

  param_codes <- str_split(raw[3], ",")[[1]] %>% str_trim()
  param_codes <- param_codes[param_codes != ""]

  col_names <- as.vector(rbind(param_codes, paste0("flag_", param_codes)))

  df <- read_csv(path, skip = 4, col_names = FALSE, show_col_types = FALSE,
    na = c("", "NA"))
  df <- df[, seq_along(col_names)]
  names(df) <- col_names

  # Pull a column by name if it exists, otherwise return a column of NA
  get_col <- function(df, name) {
    if (name %in% names(df)) df[[name]] else rep(NA_real_, nrow(df))
  }

  tibble(
    depth = get_col(df, "PRESR01"),
    temperature = get_col(df, "TEMPST01"),
    salinity = get_col(df, "PSALST01"),
    chlorophyll = get_col(df, "CPHLPR01"),
    cruise = meta$Cruise,
    cast_id = meta$Orig_Id,
    latitude = parse_dms(meta$Latitude),
```

```

    longitude = parse_dms(meta$Longitude),
    start_date = meta$Start_Date
  )
}

```

Now we can read the data, and find where there's chl data too:

```

cast_files <- list.files(data_dir, pattern = "\\\\.csv$", full.names = TRUE, recursive = TRUE)

all_casts <- map_dfr(cast_files, read_amt_cast)

# keep casts with real (non-zero, non-NA) chlorophyll data
casts_with_chl <- all_casts %>%
  group_by(cast_id) %>%
  filter(any(!is.na(chlorophyll) & chlorophyll > 0)) %>%
  ungroup()

```

Let's check which cruises have chl profiles:

```

cruise_summary <- casts_with_chl %>%
  distinct(cruise, cast_id, latitude) %>%
  group_by(cruise) %>%
  summarise(
    n_casts = n(),
    lat_range = max(latitude) - min(latitude),
    .groups = "drop"
  ) %>%
  arrange(desc(n_casts))

cruise_summary

```

```

## # A tibble: 8 x 3
##   cruise n_casts lat_range
##   <chr>   <int>   <dbl>
## 1 AMT12     69    98.1
## 2 AMT6      50    83.5
## 3 AMT10     37    84.4
## 4 AMT8      34    97.3
## 5 AMT7      32    97.0
## 6 AMT5      31   100.
## 7 AMT6B     25    91.5
## 8 AMT1      23    99.7

```

Let's select our cruise:

```

chosen_cruise <- "AMT12" # swap in whichever cruise looks best from Step 1

casts_deep_enough <- casts_with_chl %>%
  group_by(cast_id) %>%
  filter(max(depth, na.rm = TRUE) > 500) %>%
  ungroup()

```

```
cruise_casts <- casts_deep_enough %>%
  filter(cruise == chosen_cruise) %>%
  distinct(cast_id, latitude) %>%
  arrange(latitude)

cruise_casts # inspect the actual latitude spread available
```

```
## # A tibble: 32 x 2
##   cast_id latitude
##   <chr>      <dbl>
## 1 AMT12_01   -50.2
## 2 AMT12_02   -49.8
## 3 AMT12_06   -44.3
## 4 AMT12_08   -43.2
## 5 AMT12_11   -40.2
## 6 AMT12_13   -36.9
## 7 AMT12_15   -34.1
## 8 AMT12_17   -31.8
## 9 AMT12_20   -28.8
## 10 AMT12_22  -24.9
## # i 22 more rows
```

And now we can select our profiles:

```
# pick the cast nearest each target latitude
target_lats <- c(45, 25, 0, -25, -50)

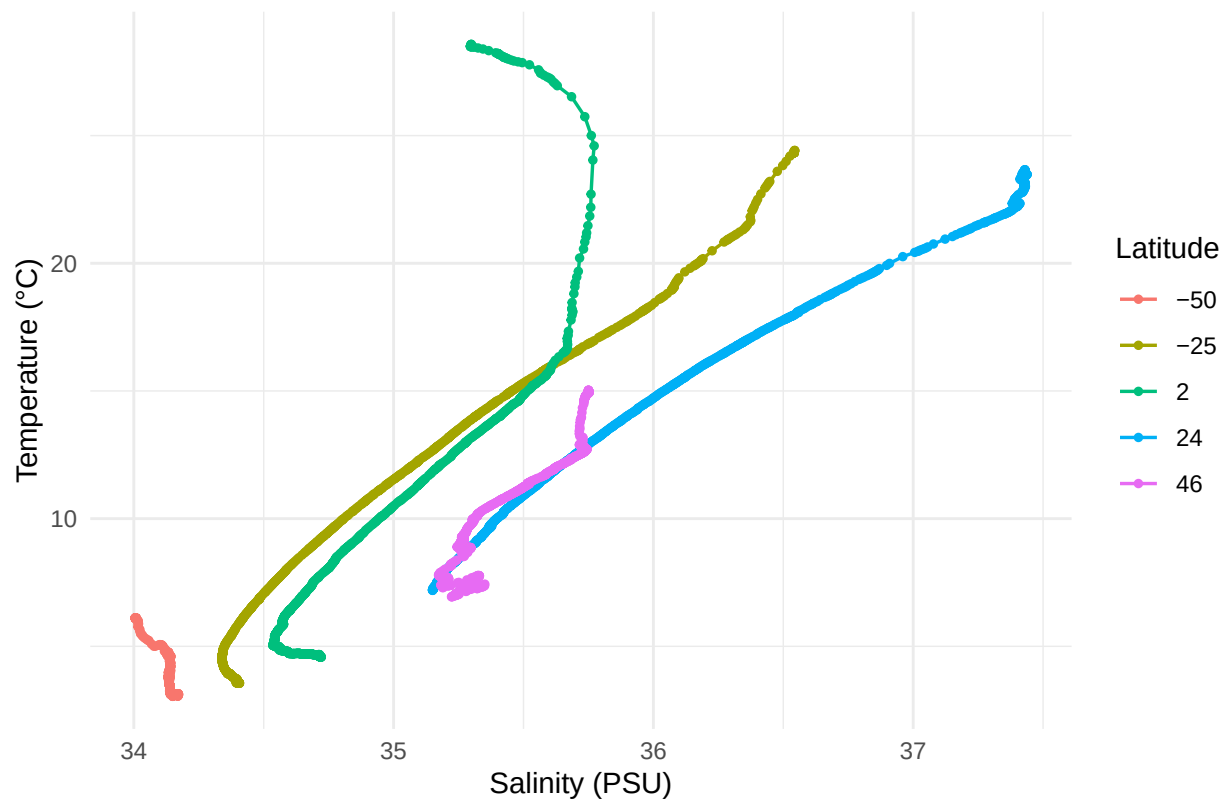
chosen_ids <- map_dfr(target_lats, function(t) {
  cruise_casts %>%
    distinct(cast_id, latitude) %>%
    mutate(dist = abs(latitude - t)) %>%
    slice_min(dist, n = 1)
})

five_profiles <- casts_with_chl %>% filter(cast_id %in% chosen_ids$cast_id)
```

And finally, plotting:

```
ggplot(five_profiles, aes(salinity, temperature, color = factor(round(latitude)))) +
  geom_path(aes(group = cast_id), linewidth = 0.6) +
  geom_point(size = 1) +
  labs(x = "Salinity (PSU)", y = "Temperature (°C)", color = "Latitude",
       title = "T-S diagram: 5 AMT profiles across the Atlantic") +
  theme_minimal()
```

T-S diagram: 5 AMT profiles across the Atlantic



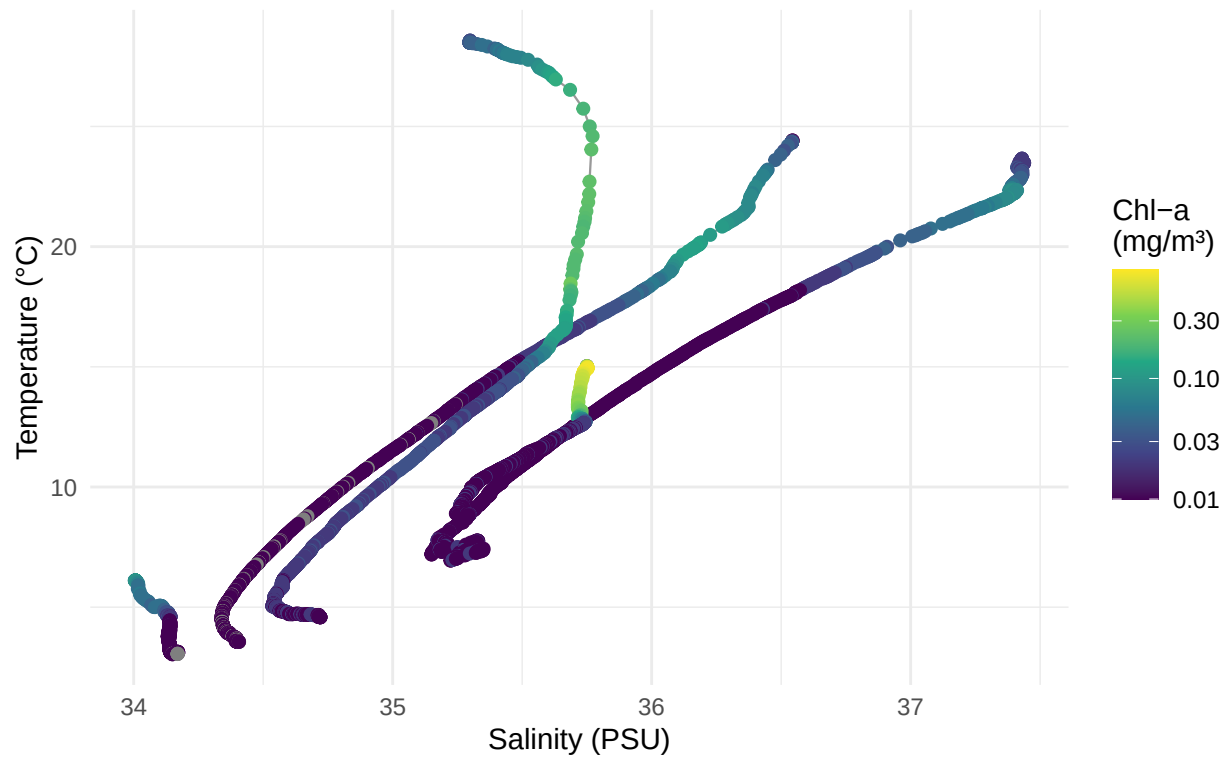
plotting by chl:

```
ggplot(five_profiles, aes(salinity, temperature)) +
  geom_path(aes(group = cast_id), color = "grey60", linewidth = 0.4) +
  geom_point(aes(color = chlorophyll), size = 1.8) +
  scale_color_viridis_c(name = "Chl-a\n(mg/m³)", trans = "log10") +
  labs(
    x = "Salinity (PSU)",
    y = "Temperature (°C)",
    title = paste("T-S diagram:", chosen_cruise, "- 5 profiles across the transect"),
    subtitle = "Points colored by chlorophyll-a concentration"
  ) +
  theme_minimal()
```

```
## Warning in transformation$transform(x): NaNs produced
```

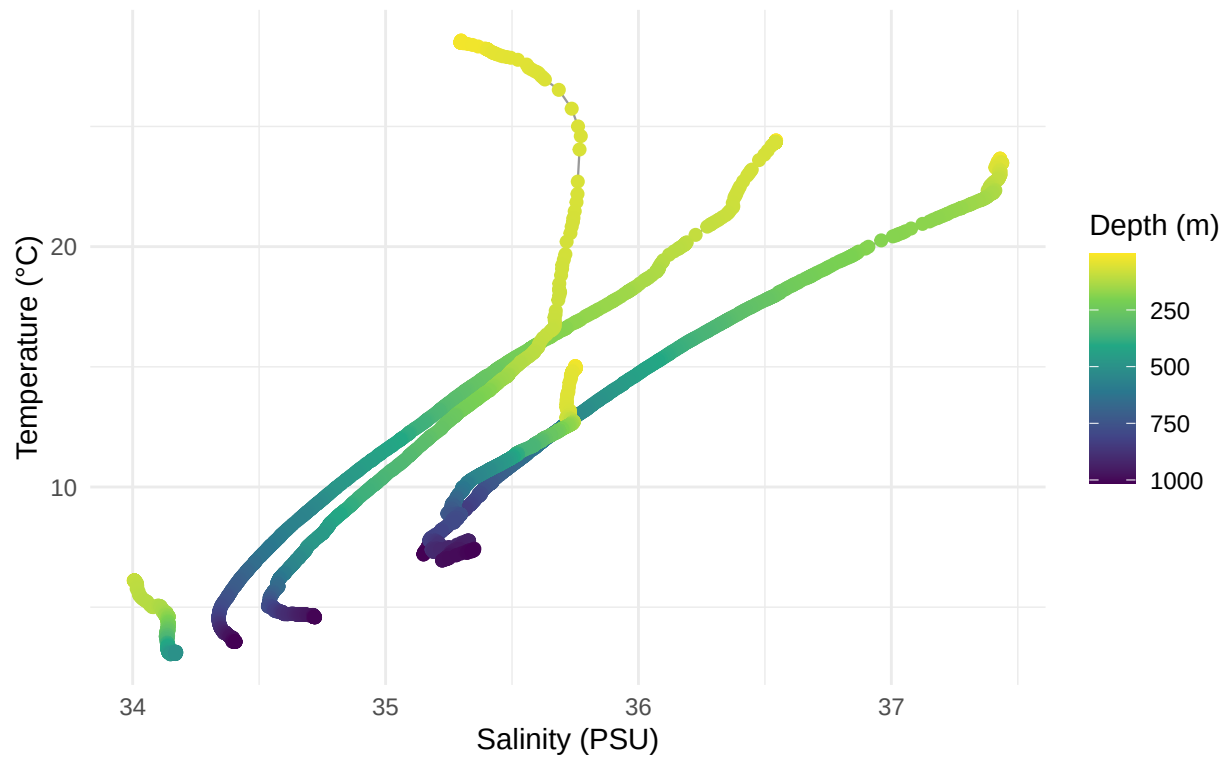
```
## Warning in scale_color_viridis_c(name = "Chl-a\n(mg/m³)", trans = "log10"):
## log-10 transformation introduced infinite values.
```

T-S diagram: AMT12 — 5 profiles across the transect
 Points colored by chlorophyll-a concentration



```
ggplot(five_profiles, aes(salinity, temperature)) +
  geom_path(aes(group = cast_id), color = "grey60", linewidth = 0.4) +
  geom_point(aes(color = depth), size = 1.8) +
  scale_color_viridis_c(name = "Depth (m)", trans = "reverse") +
  labs(
    x = "Salinity (PSU)",
    y = "Temperature (°C)",
    title = paste("T-S diagram:", chosen_cruise, "- 5 profiles across the transect"),
    subtitle = "Points colored by depth"
  ) +
  theme_minimal()
```

T-S diagram: AMT12 — 5 profiles across the transect
Points colored by depth



Saving the file:

```
write_csv(five_profiles, paste0(parent_dir, "amt_five_profiles.csv"))
```